

Etching of Silicon Nitride in CCl_2F_2 , CHF_3 , SiF_4 , and SF_6 Reactive Plasma: A Comparative Study

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Silicon nitride is an important material layer in various types of microelectronic devices. Because of continuous integration of devices, patterning of this layer requires a highly selective and anisotropic etching process. Reactive ion etching is one of the most simple and popular plasma processes. The present work is an experimental analysis of primary etch characteristics in reactive ion etching of silicon nitride using chlorine- and/or fluorine-based organic and inorganic chemistries ($\text{CCl}_2\text{F}_2 + \text{O}_2$, $\text{CHF}_3 + \text{O}_2$, $\text{SiF}_4 + \text{O}_2$, $\text{SF}_6 + \text{O}_2$, and $\text{SF}_6 + \text{He}$) in order to obtain a simultaneous etch selectivity against polysilicon and silicon dioxide. A recipe, in $\text{CCl}_2\text{F}_2/\text{O}_2$ plasma chemistry, which provides acceptable etch characteristics, along with a reasonable simultaneous selectivity against polysilicon and silicon dioxide, has been formulated.

KEY WORDS: Anisotropy; etch characteristics; isolation; selectivity; trench.

1. INTRODUCTION

Silicon nitride (Si_3N_4) has a wide variety of applications in the fabrication of silicon semiconductor devices, such as memory,⁽¹⁾ electro-optical and CMOS devices.^{2,3} One of the significant applications of silicon nitride is in masks for X-ray lithography.⁽⁴⁾ With the downward scaling of monolithic ICs, the role of silicon nitride is continuously increasing, especially in isolation techniques such as LOCOS, SWAMI, TRENCH, OSELO, and SILO.⁽⁵⁻¹⁰⁾ Therefore, the etching of Si_3N_4 needs critical control over the etch rate, selectivity⁽¹¹⁾ (against the masking and underlayers, especially thin SiO_2), and etch profile.⁽¹²⁾ Various research workers, in order to achieve the desired characteristics, have extensively studied plasma etching of this material layer. Different reactive gases investigated for this purpose include halocarbons, such as CF_4 , CHF_3 , etc, and inorganic reactive gases, NF_3 and SiF_4 , etc. Selectivity with respect to silicon dioxide in halocarbon

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plasmas had been a severe problem. A lot of effort was put in to achieve this important characteristic either through process design or reactor design, or a combination of both approaches. Freon-14 (CF_4) was found to be most economic and popular reactive gas for the plasma etching of dielectrics, including silicon nitride. In the early 1970s, Adir Jacob,⁽¹³⁾ pioneered the plasma etching of silicon nitride in CF_4 -based plasmas using a barrel reactor. Since then, much work in CF_4 -based plasma processes has been reported^(14–20) using different kinds of plasma sources (low- and high-pressure RF, ECR, TCP, HCP, ICP, etc.) and reactor configurations (hexode, barrel, tunnel, parallel plate, downstream, etc.) for the etching of silicon nitride. It has been established that a high selectivity against silicon dioxide, particularly in halocarbon plasma, is very difficult to achieve. However, it has been achieved in microwave plasma processes. Recently, Kastenmeir *et al.*⁽¹⁹⁾ have reported a chemical dry etching process for $\text{Si}_3\text{N}_4/\text{SiO}_2$ in a downstream microwave (GHz) plasma reactor using CF_4 as a reactive gas. They have achieved very good selectivity through the incorporation of O_2 and N_2 in a downstream microwave plasma process. Nitrogen plays a significant role in the increased etch rates of nitride and selectivity against silicon dioxide. They have suggested an etching mechanism to explain this phenomenon. In another work, Grykewich *et al.*⁽²⁰⁾ have reported a mask-dependent selectivity against silicon dioxide in CF_4/O_2 . A number of other halocarbons^(21–24) have also been investigated for the etching of $\text{Si}_3\text{N}_4/\text{SiO}_2$. Ying Zhang *et al.*⁽²⁴⁾ showed that the hydrogen-rich fluorocarbon gases or their mixtures, which contain less carbon e.g., CHF_3/H_2 are useful for selective etching of Si_3N_4 over SiO_2 .^(25–30) However, the selectivity against SiO_2 has been achieved through the addition of O_2 ,^{17,19,20,23,27,30,31} N_2 ,⁽¹⁹⁾ NH_3 ,⁽³¹⁾ and Cl_2 .⁽³²⁾ Most of these processes, including the microwave plasma processes, have either the disadvantage of high ratio of O_2 in the mixture, which leads to the loss of lateral dimensions of the etched patterns, or corrosion problems (for the processes with chlorine additives). Another reactive gas NF_3 ⁽³³⁾ has been reported to provide moderate selectivity for silicon nitride against silicon dioxide in RF plasma. In this process, CCl_2F_2 has been used as an additive to improve the selectivity against silicon dioxide, but the selectivity against polysilicon is doubtful. Reactive gas SiF_4 has also been reported^(34,35) to provide selectivity against silicon dioxide. However, it is not free from depositions. On the other hand, SF_6 -based plasma processes have provided a very high selectivity against silicon dioxide, but, again, the selectivity with respect to polysilicon would be a problem. Therefore, the existing RF reactive plasma processes for the etching of silicon nitride are either selective to polysilicon or to silicon dioxide. Thus, simultaneous selectivity is a problem. The present work aims at the development of a process, which provides a simultaneous selectivity against silicon dioxide and

polysilicon. We have investigated a number of chlorine- and/or fluorine-containing organic and inorganic chemistries based on four reactive gases, CCl_2F_2 , CHF_3 , SiF_4 , and SF_6 to obtain a reasonable etch selectivity against silicon dioxide as well as polysilicon, in the reactive ion etching (RIE) of Si_3N_4 . In particular, Freon-12 (CCl_2F_2), which has not, thus far, been investigated as a main reactive gas, was explored in detail in the present work for the RIE of silicon nitride. An analytical comparison of important etch characteristics have been made using five plasma chemistries and the most useful formulation has been suggested.

The characteristics of a dry etching process are numerous,⁽³⁶⁾ the most important of which are the etch rate, selectivity, and anisotropy. We would call these three parameters the primary etch characteristics. The first one governs the speed of the process and is useful in controlling throughput—a high etch rate is desirable for high throughputs. The second important characteristic is the selectivity against masking and underlayers. Without proper selectivity, correct transfer of a desired pattern onto the layer of interest cannot be achieved. The third one is the control over the etch profile. It has its implication over the feature integrity. The other etch characteristics are uniformity, reproducibility, residue or deposition-free etch, and freedom from physical and radiation damages, which can be taken care of in different ways.⁽³⁷⁾

The etch characteristics discussed above are the composite functions of a score of interdependent process parameters.⁽³⁸⁾ However, the etch chemistry is the single largest variant having the most influence compared to all other process parameters on the primary etch characteristics.⁽³⁹⁾ Therefore, the chemical composition of etch gas can be varied to obtain the best selectivity in a specific set of process parameters. In the present work, we establish that the reactive gases, viz., CCl_2F_2 , CHF_3 , SiF_4 , and SF_6 , with oxidizing and/or neutral additives, could be used to achieve a reasonably good selectivity against both SiO_2 and polysilicon in the RIE of silicon nitride.

2. EXPERIMENTAL

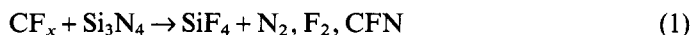
Samples used for this work consist of both patterned and unpatterned LPCVD silicon nitride, thermally grown silicon dioxide and LPCVD polysilicon on p-type $\langle 100 \rangle$ silicon wafers of 50-mm diameter. The nitride samples had an initial nitride thickness of $1500^\circ \text{\AA}$ with an underlying silicon dioxide of 1000 thickness. The vertical etch rates were determined by pre- and postetch thickness of three layers on independent wafers using the Nanospec thickness measurement system. The lateral etch rates were

deduced from the pre- and postetch linewidths of a group of lines. A high-resolution photoresist HPR 204 from Hunt Chemicals (USA) was used for patterning the samples.

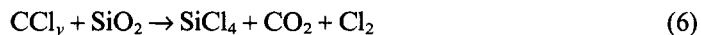
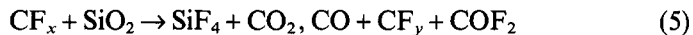
The etching experiments were conducted in Anelva DEA 506M parallel plate batch reactor. The reactor chamber is composed of 316 stainless steel. It consists of a rotating (5 RPM) water-cooled cathode and a perforated anode. The interelectrode spacing can be varied from 10 to 110 mm using spacers. Samples were placed horizontally over the cathode of a diameter of 50 cm. A Teflon cover on the cathode was used for all Freon-based plasmas whereas a quartz cover replaced it for chlorine-based plasma (CCl_2F_2) processes. All the process parameters, except the substrate temperature and the DC bias, were monitored directly. The cathode temperature was, however, maintained between 55 and 60°C in all experiments indirectly through interative control of the temperature of the cooling water circulated through the cathode. The DC bias, in fact, has a complex dependence on all other process parameters.

3. RESULTS AND DISCUSSIONS

The primary etch characteristics, viz., etch rate, selectivity against the masking and underlayer, and anisotropy in the RIE of silicon nitride are a complex function of a number of process parameters, etch chemistry being the most important among them. As already stated, five etch chemistries were chosen for an analysis of the primary etch characteristics. Through initial gauging experiments the etch pressure and interelectrode distance were optimized at 10 Pa and 70 mm, respectively. The ratio (vol.%) of the oxygen/nitrogen or helium in the reactive gas mixture was investigated (by continuously increasing its ratio in the mixture) for the optimum values of the primary etch characteristics. The gases used for CCl_2F_2 , SiF_4 , CHF_3 , and SF_6 , are either fluorine and/or chlorine based and provide F, Cl, CF_x , and CCl_y -type fragments and ions, which react with Si in Si_3N_4 to convert it to volatile products.⁽⁴⁰⁾ The possible empirical reactions are:



With polysilicon and SiO_2 , the following reactions dominate:^(41,42)



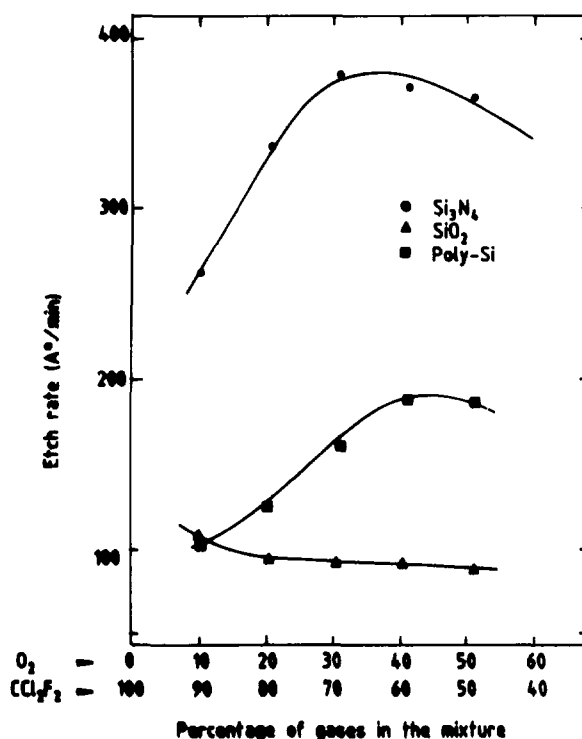


Fig. 1. Variation in the etch rates of silicon nitride, silicon dioxide, and polysilicon with the vol.% of oxygen in ($\text{CCl}_2\text{F}_2 + \text{O}_2$) reactive plasma at a total flow rate of 50 SCCM, 600 W RF power (13.56 MHz), and 10 Pa etch pressure.

There are at least five sequential primary steps in a typical RIE process.⁽⁴³⁾ The generation of reactive species essential for the etching of silicon nitride in a reactive plasma is dependent upon the RF power, flow rate, and etch pressure for a given reactor configuration. These process parameters have been fixed at certain values for all the etching recipes analyzed in order to compare the etch characteristics. Hence, the residence time for reactive species in all plasma chemistries except one ($\text{CHF}_3 + \text{O}_2$ chemistry) was fixed at 5.68 m/s. In $\text{CHF}_3 + \text{O}_2$ plasma, it was not possible to obtain a desired selectivity against silicon dioxide at the etch pressure of 10 Pa (set for all the processes). We have observed that a considerable selectivity against SiO_2 can be achieved at an etch pressure of 15 Pa and, hence, the residence time comes out to be 8.52 m/s.

One of the biggest problems with the RIE of silicon nitride is that the species reactive to silicon nitride are almost equally reactive to silicon dioxide and also, to a great extent, to polysilicon.^(38,44) Therefore, it is very difficult to devise a recipe, that provides acceptable selectivity in the reactive

plasma etching of silicon nitride against silicon dioxide and polysilicon simultaneously. Nevertheless, the present study reveals that the important plasma etch characteristics can be tailored through the judicious selection of etch chemistry and optimization of the process parameters. We will now discuss the results obtained with different plasma chemistries.

Freon-12 (CCl_2F_2) is a gas which provides both Cl and F species in its plasma and, hence, can be used for the reactive plasma etching of silicon nitride. Thus far, there is hardly any work reported using Freon-12 as a main etching gas for the etching of silicon nitride. The addition of a small percentage of gas in plasma has been reported⁽³³⁾ to enhance the selectivity of silicon nitride against silicon dioxide. A very high selectivity against silicon dioxide has been recently reported^(31,45,46) using NF_3 with $\text{O}_2/\text{NH}_3/\text{N}_2$ in microwave plasma processes. However, selectivity against polysilicon in NF_3 -based plasma is again a problem. In the present work we have devised a recipe, based in Freon-12, which is able to provide a simultaneous selectivity against both the SiO_2 and polysilicon layers. We have observed that the etch rates of silicon nitride and selectivity against SiO_2 and polysilicon are rather low in CCl_2F_2 RIE without any additive. The addition of inert additives (He, Ar) enhances etch rates but not selectivity. On the contrary, an oxidizing additive, e.g., O_2 , enhances etch rate and selectivity against SiO_2 and polysilicon. Figure 1 shows the variation in etch rates of the three material layers: Si_3N_4 , SiO_2 , and polysilicon with the variation of vol.% of O_2 in the mixture. We observe that the etch rates are lower for both silicon nitride and silicon dioxide, with the lower ratios of oxygen in the mixture. With the increase of O_2 , a rise in etch rates is observed. Empirically, CCl_2F_2 breaks into a number of fragments in plasma:



The initial low etch rates at lower ratios of oxygen in the mixture can be attributed to the lower steady state concentrations of F and Cl species because of their recombination with the carbon atoms and unsaturated species, as is clear from Eq. (8). With an increase of oxygen in the mixture, the carbon atoms are converted to CO or CO_2 , as shown in Eq. (9). As an outcome, the recombination of F and Cl species with carbon decreases leading to an increase in the number of etch species. This results in an increase in etch rates. This increase in etch rates of Si_3N_4 , however, stops at 35-vol.% of O_2 in the mixture. We reach a maximum etch rate of $378 \text{ \AA}/\text{min}$. Obviously, the further generation of reactive species is replaced by oxygen atoms of the feed gas. If we further increase the ratio of O_2 in the mixture, it keeps reducing F and Cl concentrations and, thereby, a reduction in etch

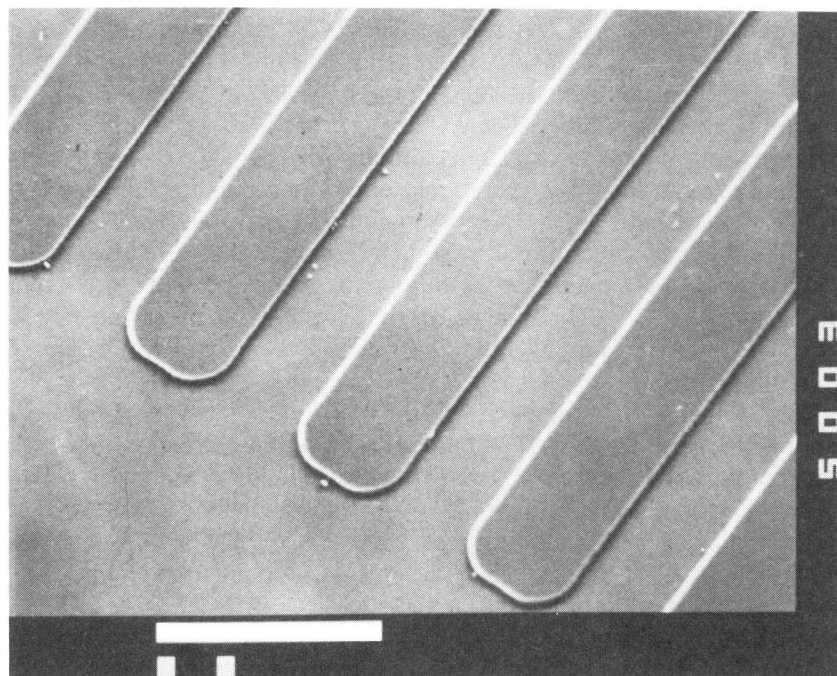


Fig. 2. SEM micrograph of a group of equidistant lines etched in 40 SCCM CCl_2F_2 + 10 SCCM O_2 at 10 Pa etch pressure and 600 W RF power.

rate is observed. The etch rate of polysilicon also shows a similar behavior, although the rate of increase is rather low; it attains a maximum value of $192.5 \text{ \AA}/\text{min}$ at 43% of O_2 . With this behavior, the selectivity of Si_3N_4 against polysilicon increases with the addition of O_2 up to 35% in the mixture. The SiO_2 etch rates in this chemistry are rather low and continuously decrease with increase in the ratio of O_2 . Therefore, as O_2 is added, the selectivity against SiO_2 increases at a faster rate in comparison to the increase in selectivity against polysilicon.

We observe that the peaks of etch rates in this chemistry for Si_3N_4 and for polysilicon do not coincide. These observations can be explained by considering the plasma and the surface chemistry involved. Oxygen plays a significant role. Initially, for the lower ratios of O_2 in the mixture, it is not available for physi- and/or chemisorption on etch surface. It is more or less consumed completely by the C atoms present in the plasma, resulting in volatile products CO and CO_2 . As already stated, we observe a continuous increase in etch rates with the increase of O_2 . If O_2 is increased beyond its

consumption by C atoms, it becomes available for absorption on the different surfaces present, the etch surface being one of them. Because of this behavior, the maximum etch rate for Si_3N_4 occurs at 35% of O_2 in the mixture. In the case of polysilicon, though, it has a larger affinity with O_2 in comparison to that of Si_3N_4 , yet the absorbed O_2 moves down to the intergrain regions, leaving a very thin layer of O_2 on the top. This thin layer is vulnerable to ion bombardments, maintaining access of etch species to the bulk. A greater amount of O_2 is, therefore, required to retard the flow of etch species to the bulk. Hence, the peak in etch rate appears at a higher ratio i.e., 43% of O_2 in the mixture. In the case of SiO_2 , a slow decrease in etch rates with increase in O_2 in the mixture may be attributed to the dilution of etch species.

The etch uniformity in this chemistry was found to be very good (better than 97% over a wafer of 50-mm diameter) at a ratio of 25 to 33% of oxygen in the mixture. It decreases very sharply with the increase in oxygen beyond 33 vol.% in the mixture. At 50 vol.% of O_2 , the selectivity decreases to 78%. The percentage loss of linewidth for 1500 Å thick silicon nitride is 6%; hence, the lateral loss of the etched patterns is quite low. Figure 2 is a SEM micrograph of a group of lines etched in 40 SCCM CCl_2F_2 , 10 SCCM O_2 at 10 Pa etch pressure and 600 W RF power. We observe that the surface is clean and smooth. There is no visible physical damage to the etched surface. The various process and characterization parameters related to this process are tabulated in Table I.

Freon-23 (CHF_3) is a reactive gas widely used for the etching of silicon dioxide.^{25,26} It can also be used for the etching of silicon nitride through judicious control of process parameters,⁽²⁵⁻³⁰⁾ particularly, the chemistry and etch pressure. We have observed that although it gives a poor selectivity against SiO_2 in a particular range of the etch pressure (8–12 Pa), a reasonable selectivity has been obtained above a certain critical pressure. It supports the findings by Dulak *et al.*⁽⁴⁷⁾ $\text{CHF}_3 + \text{O}_2$ plasma has a behavior quite similar to $\text{CF}_4 + \text{O}_2$ plasma in the etching of $\text{Si}_3\text{N}_4/\text{SiO}_2$. Many of the researchers have investigated CF_4 chemistry in detail in an effort to obtain suitable primary etch characteristics for these two materials. A high selectivity against silicon dioxide in microwave plasma has been reported,⁽¹⁹⁾ which is rather difficult to obtain in RF plasma. However, the mixture of halocarbons with reducing or oxidizing additives has been reported to provide a reasonable selectivity against silicon dioxide. Most of these processes have a disadvantage of high ratio of O_2 in the mixture, which leads to the loss of lateral dimensions of the etched patterns. We have, in the present work, chosen CHF_3 , which can deliver reasonable simultaneous selectivity against SiO_2 and polysilicon, respectively, along with anisotropy. The, addition of N_2 , in a subsequent experiment, has shown only a moderate

Table I. Process and Characterization Parameters for Different Chemistries in the Reactive Ion Etching of Silicon Nitride

No.	Chemistry	Process parameters			Etch rates of Si ₃ N ₄				Vertical etch rates			Selectivity against			Loss of linewidth for 1500 Å thick Si ₃ N ₄ (%)	
		Etch pressure (Pa)	Power (Watt)	Inter-electrode distance (mm)	Typical flow rates Gas	SCCM	Vertical (Å/min)	Lateral (Å/min)	Photo resist (Å/min)	SiO ₂ (Å/min)	Poly Si (Å/min)	PR	SiO ₂	Poly Si	Si	Uniformity (%)
1.	CCl ₂ F ₂ + O ₂	10	600	70	CCl ₂ F ₂	40	340	57	204	47	117	1.7	7.2	2.9	6.0	97
2.	CHF ₄ + O ₂	15	600	70	CHF ₃	40	768	312	192	449	87	4.0	1.7	8.8	13.4	86
3.	SiF ₄ + O ₂	10	600	70	SiF ₄	40	430	333	67	170	103	6.4	2.5	4.2	16.4	—
4.	SF ₆ + O ₂	10	600	70	SF ₆	40	939	331	494	65	—	1.9	14.4	—	11	81
5.	SF ₆ + He	10	600	70	SF ₆	40	800	232	368	65	—	2.2	12.3	—	9.9	85

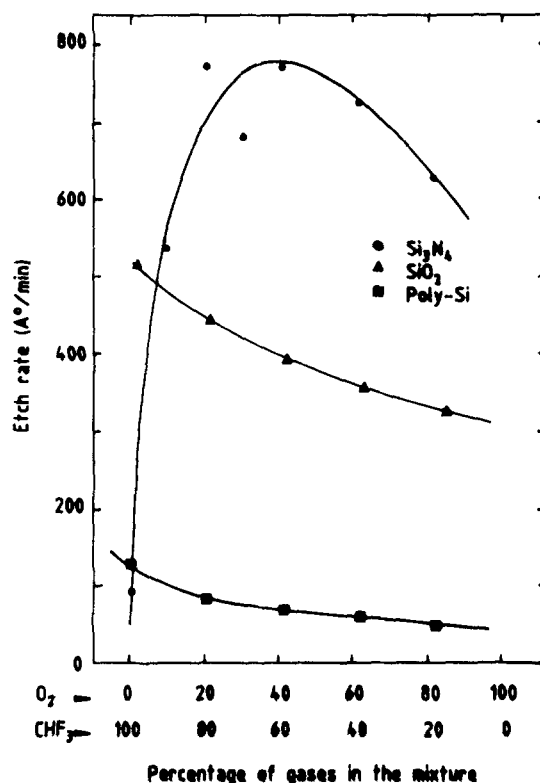


Fig. 3. Variation in the etch rates of silicon nitride, silicon dioxide, and polysilicon with the vol.% oxygen in ($\text{CHF}_3 + \text{O}_2$) reactive plasma at a total flow rate of 50 SCCM, 600 W RF power (13.56 MHz), and 15 Pa etch pressure.

improvement in the selectivity (increases to 3.1) against SiO_2 in our process. Figure 3 shows the variation in etch rates of three material layers, viz., Si_3N_4 , SiO_2 , and polysilicon with the increase in vol.% of O_2 in the mixture of CHF_3 and O_2 as reactive gas at an etch pressure of 15 Pa and RF power of 600 W. It is observed that etch rate and selectivity for the etching of Si_3N_4 are rather low for very low concentrations of O_2 in the mixture, supporting findings by some of the researchers.^(17,19,20,23,27,30) With the increase of oxygen, the Si_3N_4 etch rate initially rises to a maximum of 776 Å/min at 40% of O_2 in the mixture. The selectivity against polysilicon and SiO_2 also attain maximum values of 11.0 and 1.9, respectively. A further increase in selectivity up to 3.1 has been achieved in a subsequent experiment through 12% incorporation of nitrogen while retaining the volume ratio of CHF_3 to O_2 equal to 4:1.

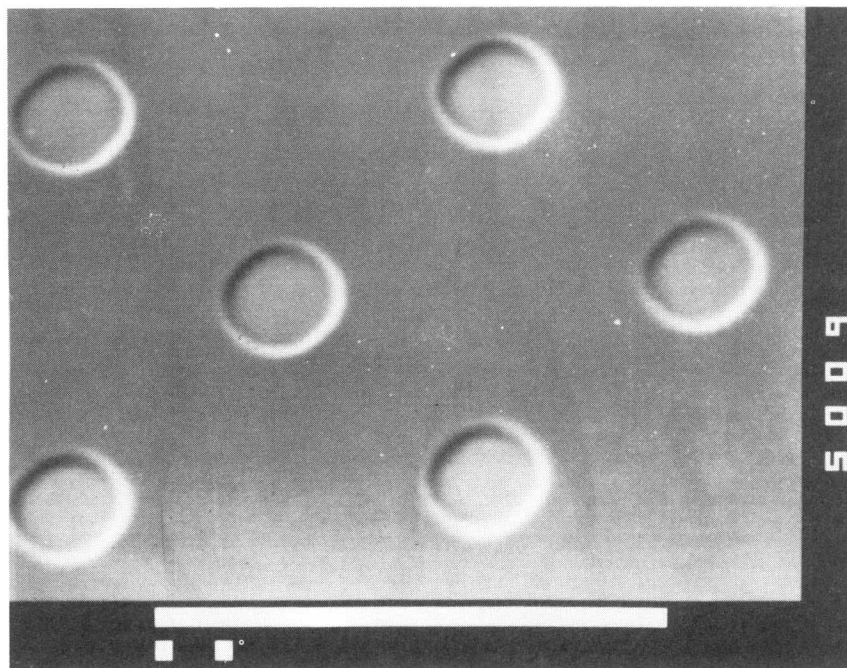


Fig. 4. SEM micrograph of 2×2 micron contact holes etched in nitride with 40 SCCM CHF_3 + 10 SCCM O_2 at an etch pressure of 15 Pa and 600 W RF power. The top and bottom surfaces are virtually free from any type of deposition and physical damage.

The rapid increase in etch rates of Si_3N_4 can be understood through the role of O_2 in the mixture. Initially, when there is no oxygen, the recombination of different species and fragments results in lower steady state concentrations of the etch species, in addition to the formation of thin polymer layers of unsaturated fluorocarbon fragments on Si_3N_4 , resulting in lower etch rates. As soon as oxygen is introduced, it oxidizes the fluorocarbon fragments, which ultimately results in a considerable increase in etch species. It also forms volatile oxides of nitrogen present in the silicon nitride. The formation of nitric oxide (NO) in the plasma has direct relation to the Si_3N_4 etch rate. It has recently been established⁽¹⁹⁾ that NO molecule or metastable NO plays a significant role in the etching of Si_3N_4 through active surface reactions. In addition to these, the fluorocarbon polymer film, which is deposited over Si_3N_4 during oxygen-deficient plasma, is decomposed quickly and is evaporated as CO or CO_2 . The total effect is a rapid increase in the etch rates. The fall in etch rates beyond 40% of O_2 in the mixture is basically because of the dilution of the reactive species by oxygen itself. The effect of

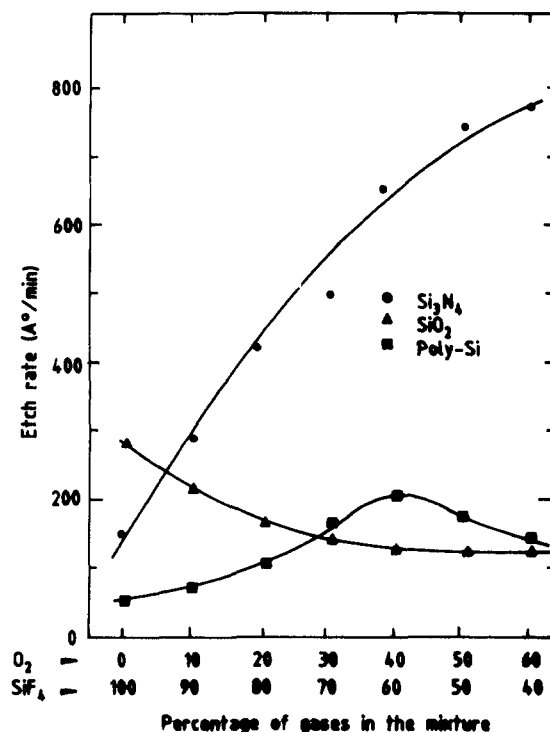


Fig. 5. Variation in the etch rates of silicon nitride, silicon dioxide, and polysilicon with the vol.% of oxygen in ($\text{SiF}_4 + \text{O}_2$) reactive plasma at a total flow rate of 50 SCCM, 600 W RF power (13.56 MHz), and 10 Pa etch pressure.

increase of oxygen in the mixture onto the silicon dioxide and polysilicon etch rates is not very significant, both of them demonstrate a slow decrease. Etch uniformity obtained in this recipe is reasonably good (86%). The process and characterization parameters for a typical process comprised of 15 Pa etch pressure, 600 W RF power, and with a total flow of 50 SCCM in CHF_3/O_2 has been given in Table I. Figure 4 is a SEM micrograph of $2 \times 2 \mu\text{m}$ contact holes etched in nitride with 40 SCCM $\text{CHF}_3 + 10$ SCCM O_2 at an etch pressure of 15 Pa and 600 W RF power. The top and bottom surfaces are virtually free from any kind of deposition and physical damage.

Silicon tetrafluoride (SiF_4) is an inorganic reactive gas, which could be an alternate to the ozone-depleting freons for the etching of dielectrics, including Si_3N_4 . Only a few references^(34,35) reporting its use in the etching of Si_3N_4 are available. In our work we observe that the variations in Si_3N_4 etch rate in $\text{SiF}_4 + \text{O}_2$ chemistry is rather unusual. As shown in Fig. 5, the etch rates keep on increasing with the addition of O_2 . The selectivity against

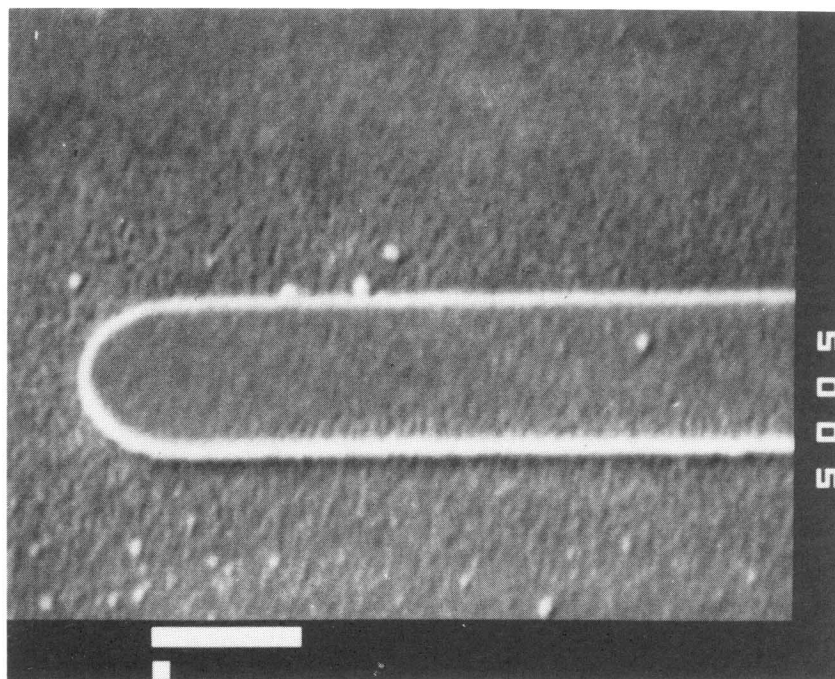


Fig. 6. One micron line etched in 40 SCCM SiF_4 + 10 SCCM O_2 at a total flow rate of 150 SCCM, 600 W RF power, and 10 Pa etch pressure. A polymer type deposition was observed.

photoresist is 6.4, which is reasonably good. A polymer-type deposition was observed in this process (Fig. 6). It is because of the deposition of the unsaturated molecules forming a thin film over silicon and silicon nitride surfaces. Photoresist nibbling, up to some extent, was also observed in the process. The etch rates are rather low for lower concentrations of O_2 in the mixture. However, the etch rates shoot up sharply with the increase of oxygen in the mixture. The selectivity against polysilicon also increases as the increase in the etch rates of polysilicon is much slower compared to that of silicon nitride. Selectivity against silicon dioxide increases sharply beyond 8% O_2 in the mixture. Selectivity against photoresist is also reasonably good (typical value 6.4), but the lateral loss of the etched patterns is higher—16.4% for 1500 Å thick nitride.

Sulfur hexafluoride (SF_6) has been the conventional reactive gas for the RIE of polysilicon/Si.^(48–51) It can, however, be used for the RIE of silicon nitride. Therefore, selectivity for silicon nitride against polysilicon in SF_6 + O_2 plasma is a problem. We have carried out a detailed investigation of the etch rates and selectivity for the silicon nitride. The selectivity against

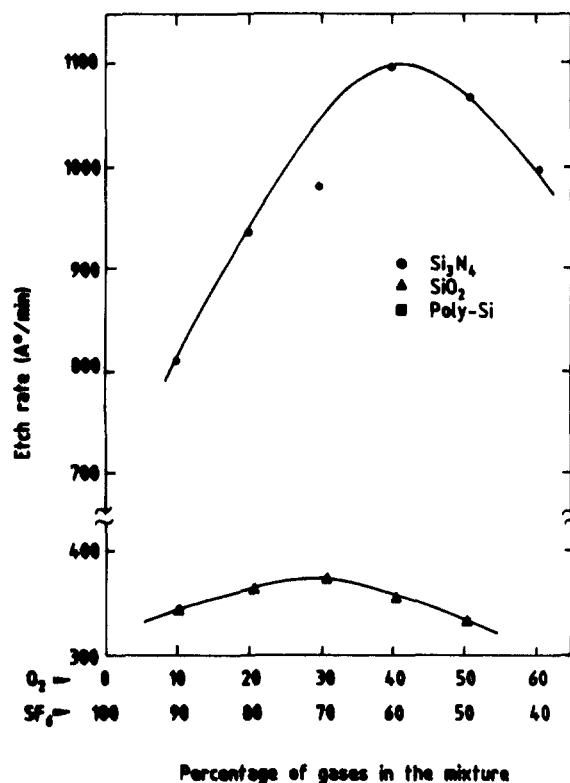


Fig. 7. Variation in the etch rates of silicon nitride and silicon dioxide with the vol.% of oxygen in ($SF_6 + O_2$) reactive plasma at a total flow rate of 50 SCCM, 600 W RF power (13.56 MHz), and 10 Pa etch pressure.

SiO_2 is reasonably good in this chemistry, caused by high concentrations of F atoms in the plasma. For the lower ratios of oxygen, F atom concentration is limited by its recombination with S atoms. An increase of O_2 in the mixture removes the S atoms in the form of volatile SO_2 , increasing the F atom concentration and, in turn, the etch rate of Si_3N_4 . Figure 7 shows the variation in etch rates of silicon nitride and silicon dioxide with the increase of oxygen in the mixture. It is observed that the Si_3N_4 etch rates increase very sharply up to 30% O_2 , and tend to stabilize between 30 to 40%. On the contrary, the SiO_2 etch rates are more or less constant, a steady increase in the etch rates up to 30% of oxygen and then a fall is observed. The selectivity against SiO_2 increases slowly with the addition of O_2 up to 30% O_2 in the mixture, whereas between 30 to 40% O_2 , the increase in selectivity is very fast because of the decrease in SiO_2 etch rates beyond 30 vol.% of O_2 in the mixture. One of the problems in this plasma process is a

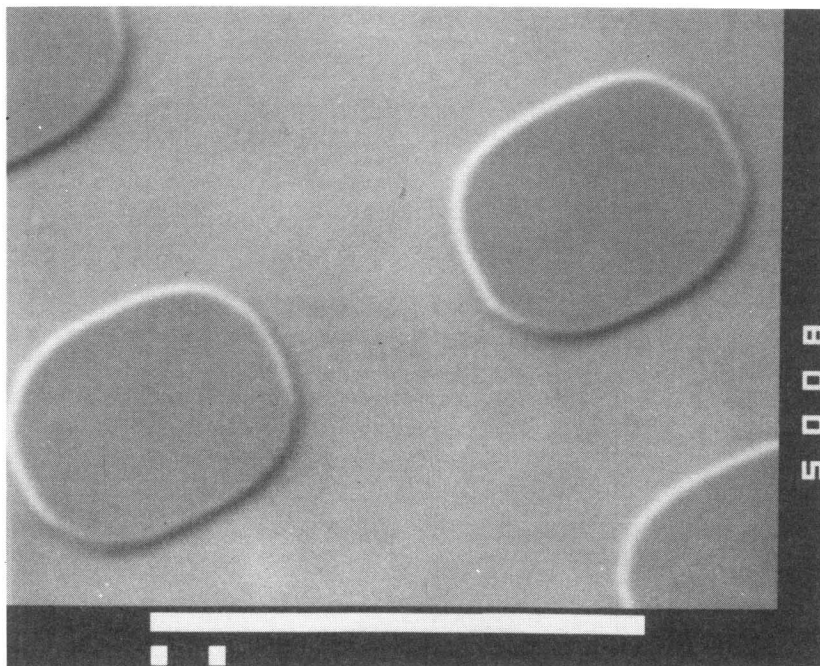


Fig. 8. Mesa structures of dimensions $6 \times 6 \mu\text{m}$ etched in a recipe comprised of 40 SCCM SF_6 + 10 SCCM O_2 at an RF power 600 W and etch pressure 10 Pa.

lower selectivity against photoresist, the best being 1.9. Obviously, it decreases with the addition of oxygen in the mixture. SF_6/O_2 provides a very clean etch, although somewhat nonuniform (81% uniformity). Figure 8 shows a few mesa structures of dimensions $6 \times 6 \mu\text{m}$ etched in 40 SCCM SF_6 + 10 SCCM O_2 , RF power of 600 W, and etch pressure 10 Pa. Introduction of an inert gas (helium) in place of oxygen in the reactive gas mixture was found to predictably contribute to the sputter etch section in the RIE. The initial low etch rates ($600 \text{ \AA}/\text{min}$, Fig. 9) at lower ratios of He in the ($\text{SF}_6 + \text{He}$) mixture are improved to $1000 \text{ \AA}/\text{min}$, because of the sputter etch mechanism of the RIE process. The process provides more anisotropic etching with a likelihood of electrical damage to the underlayer, which could be minimized through the judicious selection of the RF power density. In another approach one can choose a two-step plasma process. The first one comprised of a highly anisotropic step, while the second one highly selective to the underlayer. Selectivity against silicon dioxide is 12.3, which is slightly lower than that obtained in SF_6/O_2 plasma chemistry. In this process, selectivity against the photoresist is improved to 2.2 and the loss of lateral dimensions is reduced to 9.9% for $1500 \text{ \AA}$ thick nitride patterns. Figure 10 is a

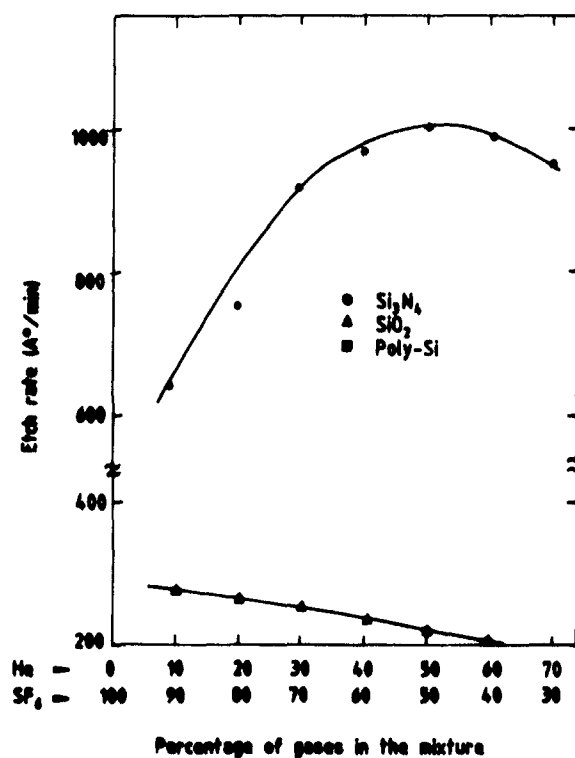


Fig. 9. Variation in the etch rates of silicon nitride and silicon dioxide with the vol.% of helium in (SF₆ + He) reactive plasma at a total flow rate of 50 SCCM, 600 W RF power (13.56 MHz), and 10 Pa etch pressure.

micrograph of $2 \times 2 \mu\text{m}$ holes etched in silicon nitride using an SF₆/He plasma with a total flow of 50 SCCM, 600 W RF power, and 10 Pa etch pressure.

4. CONCLUSIONS

A comparison of five RIE processes based on CCl₂F₂, CHF₃, SiF₄ and SF₆ reactive RF (13.56 MHz) plasma for the etching of silicon nitride has been made. The typical process parameters and the etch characteristics (etch rates, selectivity against the photoresist, silicon dioxide, polysilicon, the percentage loss of linewidth of the patterns of thickness 1500 Å in silicon nitride and uniformity) obtained for the five recipes are summarized in Table 1.

It has been observed that oxygen addition has a significant effect in the improvement of primary etch characteristics of silicon nitride. In RF

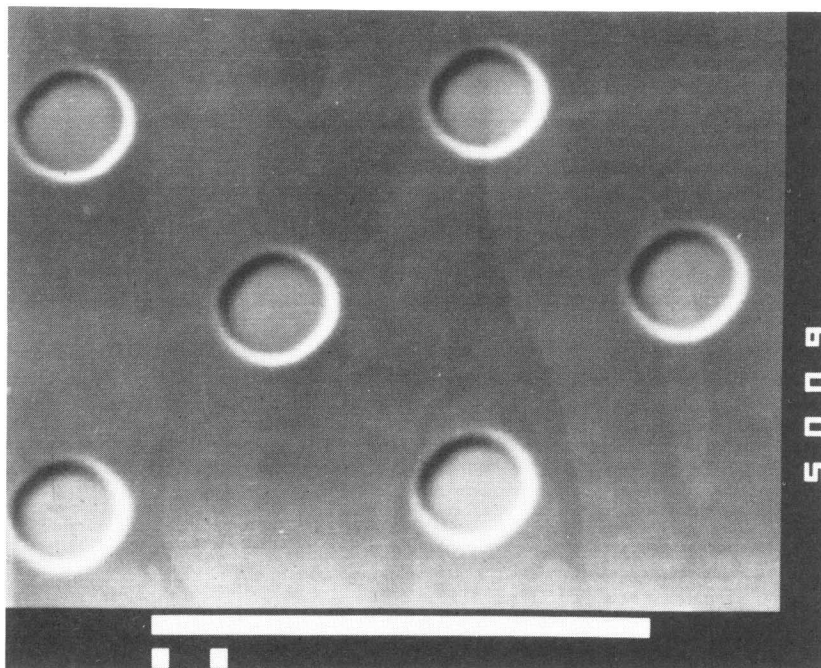


Fig. 10. Micrograph of $2 \times 2 \mu\text{m}$ holes etched in silicon nitride using an SF_6/He plasma with a total flow of 50 SCCM, 600 W RF power, and 10 Pa etch pressure.

(13.56 MHz) plasma processes, we have observed that $\text{CCl}_2\text{F}_2/\text{O}_2$ and CHF_3/O_2 plasma based RIE processes are capable of delivering a reasonable simultaneous selectivity against polysilicon and silicon oxide, yet the earlier one has a clear edge in anisotropy and deposition-free etch surfaces. The later one, i.e., CHF_3/O_2 could be used effectively in the fabrication processes where one can sacrifice a marginal loss of the lateral dimension of the etched patterns. It provides a relatively higher selectivity against polysilicon. For a highly anisotropic etching one, can use SF_6/O_2 selectively against silicon dioxide. More effectively, one can choose a two-step process based on two different plasma chemistries, depending upon a particular requirement.

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